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REVIEW ARTICLE

The effect of indoor air filtration on biomarkers of inflammation and oxidative stress: a review and meta-analysis of randomized controlled trials

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Abstract

Improvement of indoor air quality is beneficial for human health. However, previous studies have not reached consistent conclusions regarding the effects of indoor air filtration on inflammation and oxidative stress. This study aims to determine the relationship between indoor air filtration and inflammation and oxidative stress biomarkers. We conducted an electronic search that evaluated the association of indoor air filtration with biomarkers of inflammation and oxidative stress in five databases (PubMed, Cochrane Library, EMBASE, Web of Science, and Scopus) from the beginning to April 23, 2023. Outcomes included the following markers: interleukin-6 (IL-6), tumor necrosis factor-α (TNF-α), C-reactive protein (CRP), malondialdehyde (MDA), 8-hydroxy-2deoxyguanosine (8-OHdG), and 8-iso-prostaglandinF2α (8-isoPGF2α). We extracted data from the included studies according to the system evaluation and the preferred reporting item for meta-analysis (PRISMA) guidelines and used the Cochrane risk of bias tool to assess bias risk. Our meta-analysis included 15 studies with 678 participants to assess the combined effect size. The meta-analysis demonstrated that indoor air filtration could have a marked reduction in IL-6 (SMD: -0.275, 95% CI: –0.545 to –0.005, $p = 0.046$) but had no significant effect on other markers of inflammation or oxidative stress. Subgroup analysis results demonstrated a significant reduction in 8-OHdG levels in the subgroup with < 1 day of duration (SMD: –0.916, 95% CI: –1.513 to –0.320; $p = 0.003$) and using filtrete air filter (SMD: –5.530, 95% CI: –5.962 to –5.099; $p < 0.001$). Our meta-analysis results depicted that indoor air filtration can significantly reduce levels of inflammation and oxidative stress markers. Considering the adverse effects of air pollution on human health, our study provides powerful evidence for applying indoor air filtration to heavy atmospheric pollution.

Keywords Indoor air filtration · Inflammation · Oxidative stress · Review · Meta-analysis · Randomized controlled trials

Introduction

With rapid global urbanization and industrialization development, air pollution has seriously threatened public health (Cohen et al. 2017). According to the Global Burden of Diseases (GBD) Study 2017, 4.90 million deaths and 1.47 billion disability-adjusted life years are

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attributed to air pollution (Collaborators 2018). Epidemiological studies have pointed out that short- and long-term exposure to ambient air pollution increases the morbidity and mortality of the population (Franchini and Mannucci 2009). Since individuals spend most of their time indoors (Tamerius et al. 2013), improving indoor air quality may be a potentially effective prevention strategy.

Much evidence indicates that short- and long-term indoor air filtration can significantly reduce indoor $PM_{2.5}$ concentrations and potentially benefit cardiovascular and respiratory systems (McDonald et al. 2002; Walzer et al. 2020). Allen et al. (R. W. Allen et al. 2011) conducted a randomized crossover intervention study of 45 healthy adults affected by wood smoke. The participants were exposed to filtered and unfiltered air for seven consecutive days. The results revealed that indoor air filtration was associated with improved endothelial function and reduced inflammatory biomarkers (Song et al. 2019). Additionally, some recent studies have demonstrated that indoor air filtration can improve cardiopulmonary health of the people (Walzer et al. 2020; Wang et al. 2021).

Many studies have explored the link between indoor air filtration, oxidative stress, and inflammation. Several studies have reported that indoor air filtration has potential health benefits in air-polluted weather by reducing inflammation and oxidative stress (Ren et al. 2022; Zhao et al. 2020), indicating that indoor air filtration may be an effective measure to prevent cardiorespiratory damage caused by air pollution. However, other investigators have concluded the opposite. Shao et al. (Shao et al. 2017) conducted a randomized crossover intervention trial in 35 non-smoking senior participants with and without chronic obstructive pulmonary disease. The research findings indicated that indoor air filtration can significantly improve indoor air quality but has little effect on systemic oxidative stress and inflammation. There were no demonstrable changes in the cardiorespiratory outcomes of older adults. In another study, indoor air filtration was associated with improved endothelial function and lower concentrations of inflammatory biomarkers but not with markers of oxidative stress (R. W. Allen et al. 2011). Accordingly, the effects of indoor air filtration on oxidative stress and inflammation remain unclear. Because oxidative stress and inflammation are closely related to the occurrence and development of many diseases, it is particularly important to clarify the impact of indoor air filtration on oxidative stress and inflammation to explore the benefits of indoor air filtration in air pollution weather. Therefore, to provide healthcare professionals with more comprehensive and accurate evidence of the effects of indoor air filtration, this systematic meta-analysis was conducted to determine the effects of indoor air filtration on biomarkers of inflammation and oxidative stress.

Materials and methods

This research was based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) criteria (Page et al. 2021) and registered in the International System Review Forward Looking Register (Prospero) database with registration number CRD42022316442.

Search strategy

Five databases (PubMed, Cochrane Library, EMBASE, Web of Science, and Scopus) were searched electronically from their inception to April 23, 2023, without restrictions on the search, including language and publication period. Searches were re-run just before the final analyses, and further studies were retrieved for inclusion. Moreover, additional RCTs were identified by manually searching the reference lists of relevant reviews and articles (Supplementary Material Table S1). The retrieved manuscripts were imported into Endnote software (version × 9), filtered and proposed the process, and duplicate manuscripts were deleted. The inclusion and exclusion criteria are presented in Table 1. The two authors (ZAZ) and (QQC) independently and cooperatively determined the manuscripts suitable for inclusion. After screening, they were all submitted to the third author (JYR), who made a summary. Studies with different opinions were decided by discussion.

Data extraction

Data were manually extracted by two independent authors and stored in a spreadsheet. The following information was collected: first author, year, published journal, inclusion

Table 1 PICOS criteria for inclusion of studies

Parameter	Description
Population	Adult participants (age ≥ 18 years)
Intervention	Indoor air filtration
Comparator	Placebo filtration, or sham air filter
Outcomes	Outcomes regarding at least one of the following markers:
	Interleukin-6 (IL-6)
	Tumor Necrosis Factor-α (TNF-α)
	C- reactive protein (CRP)
	Malondialdehyde (MDA)
	8-hydroxy-2 deoxyguanosine (8-OHdG)
	8-iso-prostaglandinF2α (8-isoPGF2α)
Study design	Randomized placebo-controlled clinical trial with a cross-over or parallel design

criteria, intervention site, filtration technology, relevant characteristics of participants (age, BMI, male (M/F), participants (n), and health status), and other relevant characteristics of interest. The data are presented as mean ± SD.

Assessment of quality

The risk of bias in qualified studies was assessed according to the Cochrane criteria (Higgins et al. 2011). Bias risk items were as follows: evidence generated by random sequence, allocation concealment, blind method, incomplete outcome data, selective reporting, and other potential sources of bias. The results of the assessment applied to data synthesis. Three reviewers were involved in the quality assessment. Disagreements between reviewer judgements will be resolved through further discussion.

Quantitative data synthesis

All data analyses were performed using Stata version 11, using STD Mean Difference (SMD), SD, and 95% Cl as measures of IL6, TNF-α, CRP, MDA, 8-OHdG, and 8-isoPGF2α. The net change was equal to the value after intervention minus the baseline value. Because some published articles did not provide baseline values, post-intervention values were used for analysis. The SD formula for the average value is as follows:

$$\mathrm{SD} = \text{square root}\left[(\mathrm{SDpre} - \mathrm{treatment})^2 + (\mathrm{SDpre} - \mathrm{treatment})^2 - (2 * \mathrm{R} * \mathrm{SDpre} - \mathrm{treatment} * \mathrm{SDpost} - \mathrm{treatment})\right]$$

Assuming a correlation coefficient (R) of 0.5 for the crossover design trial, we used the heterogeneity index I^2 to assess heterogeneity between different studies. In each study, if $I^2 < 50\%$, indicating no statistical heterogeneity, using the fixed effect model, if $I^2 > 50\%$, a random-effects model will be used. To evaluate the impact of individual studies on the size of an aggregate effect, a sensitivity analysis was conducted, and $p < 0.05$ was considered statistically significant.

Subgroup analysis

Subgroup analysis was performed according to intervention duration (< 1 day or > 1 day), Filtration technology (filtrete air filter or HEPA filter) (Zhu et al. 2021), and BMI (< 24 kg/m^2 or > 24 kg/m^2).

Publication bias

We used Begg's rank correlation test, Egger's linear regression test, and funnel plot to explore potential publication bias (Begg and Mazumdar 1994; Egger et al. 1997) and it was considered statistically significant when p value < 0.1.

Results

Database retrieval process and characteristics of inclusion in the study

We systematically searched the five databases on April 23, 2023. The initial search identified 677 papers to be screened, of which 125 were rejected due to duplication. After title and abstract screening, 513 papers were rejected because their inclusion criteria were inconsistent with this study. The remaining 39 papers were further screened, and 24 were rejected for the following reasons: no relevant data ($n = 10$), no RCT ($n = 6$), wrong intervention ($n = 4$), Review, News or Comment ($n = 4$). Finally, 15 studies (J. Allen et al. 2009; R. W. Allen et al. 2011; Bräuner et al. 2008; Chen et al. 2015; Chuang et al. 2017; Cliff et al. 2016; Guo et al. 2021; Kajbafzadeh et al. 2015; Liu et al. 2015; Riddervold et al. 2012; Shao et al. 2017; Törnqvist et al. 2007; Wang et al. 2021; Wen et al. 2022; Zhao et al. 2020) with 678 participants met the inclusion criteria. A PRISMA flowchart of this study is displayed in Fig. 1.

Overview of the characteristics of the included studies

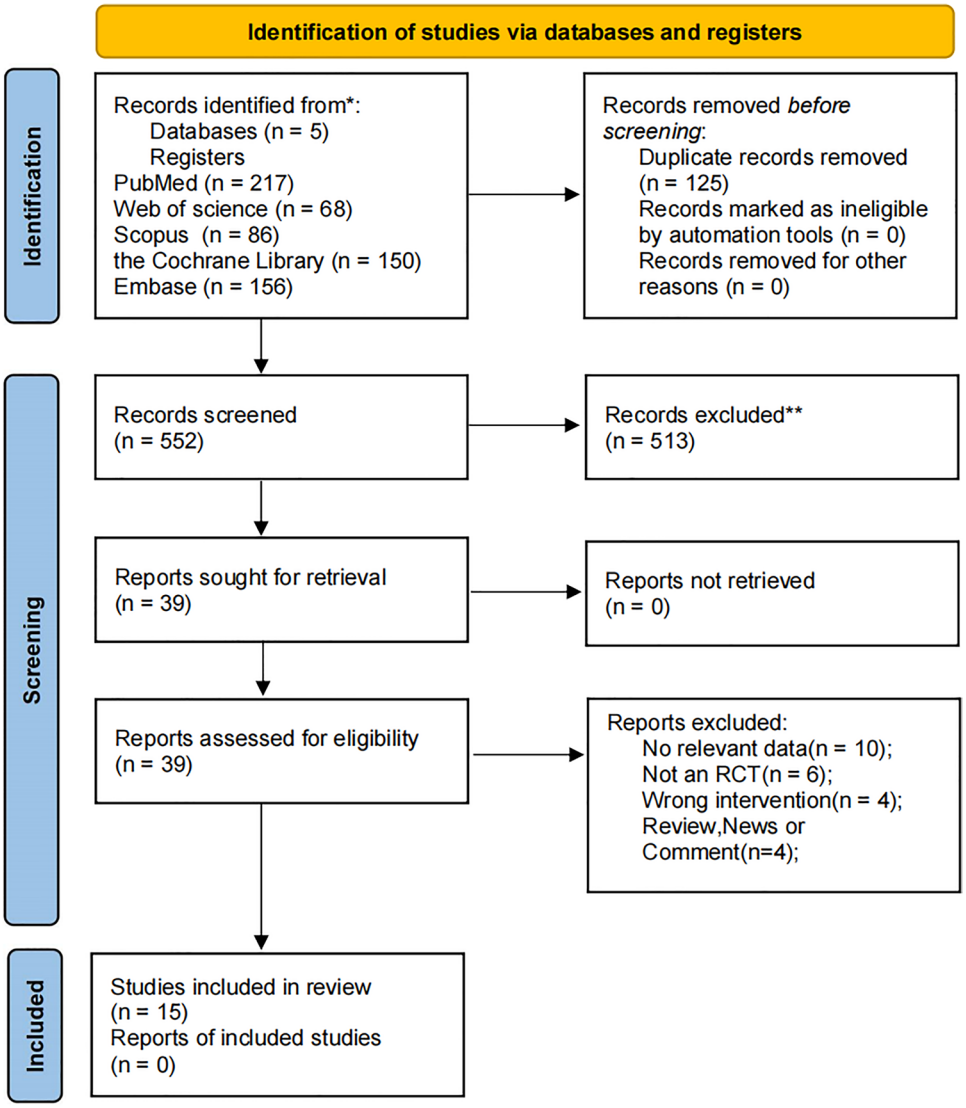
The 15 studies included were published centrally from 2007 to 2023 and conducted in the following five countries: Sweden, Denmark, the United States, Canada, and China. A total of 678 subjects were included in this study. The intervention time ranged from 1 h to 12 months. Nine studies selected HEPA filters, and the remaining six used filtered air filters. The intervention sites included the following four locations: vehicle, personal residence, laboratory, and quarter. Twelve studies contained indicators of inflammation and nine contained indicators of oxidative stress (Table 2).

Outcomes of meta-analysis

Effect of indoor air filtration on inflammation

Ten studies reported the impact of indoor air filtration on IL-6 level, eight on CRP level, and six on TNF-α level, involving 360 subjects. We used a random-effect model for IL-6 ($I^2 = 69.0\%$, $p < 0.001$; Fig. 2), TNF-α ($I^2 = 67.2\%$, $p = 0.009$; Fig. 2), and CRP ($I^2 = 96.3\%$, $p < 0.001$; Fig. 2). We found that indoor air filtration significantly reduced the levels of IL-6 (SMD: –0.275, 95% CI: –0.545 to –0.005,

Fig. 1 The PRISMA flowchart



$p = 0.046$; Fig. 2) but had no significant effect on the levels of TNF-α (SMD: –0.016, 95% CI: –0.375 to 0.342, $p = 0.929$; Fig. 2) or CRP (SMD: –0.320, 95% CI: –1.116 to 0.476, $p = 0.431$; Fig. 2).

Effect of indoor air filtration on oxidative stress

We found four studies involving 100 participants that measured the effects of indoor air filtration on MDA levels. The Combined results indicated that indoor air filtration did not affect MDA levels (SMD: –0.473, 95% CI: –1.144 to 0.197, $p = 0.166$; Fig. 2).

Five trials with 247 participants measured the effect of indoor air filtration on 8-OHdG and 8-isoPGF2α levels. The results displayed that indoor air filtration had no significant effect on the levels of 8-OHdG (SMD: –1.420, 95% CI: –3.331–0.491, $p = 0.145$; Fig. 2) or 8-isoPGF2α (SMD: 0.109, 95% CI: –0.351–0.569, $p = 0.642$; Fig. 2).

Risk-of-bias assessment

The quality of the studies was assessed using the Cochran collaboration bias risk assessment tool (Higgins et al. 2011). In all included studies, the blind methods of random sequence generation, allocation concealment, and result evaluation all had a low risk of bias. A high risk of bias only appeared in the incomplete results. The details of the study quality are illustrated in Table 3.

Subgroup analysis

We performed a subgroup analysis that included intervention duration (< 1 day vs. ≥ 1 day), filtration technology (filtrete

Table 2 Characteristics of the included trials in this meta-analysis

Reference	country	Sample size	Sex (M/F)	Mean age(y)	BMI (kg/m^2)	Intervention site	Filtration technology	Duration	Main outcomes
Törnqvis et al. (2007)	Sweden	15	15/0	26.0	ns	Vehicle	Filtrete air filter	1 h	IL6, CRP, TNF-α
Bräuner et al. (2008)	Denmark	41	21/20	67.0	25.0	Individual house	HEPA filter	2 days	IL6, CRP, TNF-α
Allen et al. (2009)	USA	10	6/4	38.9	40.3	laboratory	HEPA filter	2 h	8-OHdG, 8-isoPGF2α
Allen et al. (2011)	Canada	45	21/24	43.0	25.7	Individual house	HEPA filter	7 days	IL6, CRP, MDA, 8-isoPGF2α
Riddervold et al. (2012)	Denmark	20	10/10	25.1	ns	laboratory	Filtrete air filter	3 h	IL6, 8-isoPGF2α
Kajbafzadeh et al. (2015)	Canada	68	32/36	43.8	24.9	Individual house	HEPA filter	7 days	IL6, CRP
Liu et al. (2015)	China	35	10/25	23.0	22.0	Dormitory	Filtrete air filter	130 minutes	IL6, CRP, MDA, 8-OHdG
Chen et al. (2015)	Canada	50	29/26	28.0	23.0	Individual house	HEPA filter	2 days	IL6, CRP, TNF-α
Cliff et al. (2016)	Canada	28	15/13	28.0	23.8	Vehicle	Filtrete air filter	2 h	IL6, TNF-α
Chuang et al. (2017)	China	200	100/100	43.4	23.6	Individual house	Filtrete air filter	12 months	CRP, 8-OHdG
Shao et al. (2017)	China	15	1/14	65.9	25.7	Individual house	HEPA filter	14 days	IL6, CRP, 8-OHdG
Zhao et al. (2020)	China	29	16/13	21.8	20.7	Dormitory	HEPA filter	2 days	MDA, 8-isoPGF2α
Guo et al. (2021)	China	24	12/12	82.0	24.7	Individual house	HEPA filter	2 h	IL6, CRP, TNF-α, 8-OHdG
Wang et al. (2021)	China	54	41/13	21.57	21.28	Dormitory	HEPA filter	1 week	MDA, 8-isoPGF2α
Wen et al. (2022)	China	54	41/13	21.6	21.3	Dormitory	Filtrete air filter	1 week	TNF-α

air filter vs. HEPA filter), and BMI (< 24 kg/m^2 vs. ≥ 24 kg/m^2) (Table 4). The results demonstrated a significant reduction in 8-OHdG in the subgroup with < 1 day of duration (SMD: –0.916, 95% CI: –1.513 to –0.320, $p = 0.003$; Table 4) and using filtrete air filter (SMD: –5.530, 95% CI: –5.962 to –5.099; $p < 0.001$; Table 4).

Sensitivity analysis and publication bias

Sensitivity analysis of IL-6 indicated that removing any study at a time had no significant effect on the pooled results (Figure S1). Sensitivity analyses of the other indicators demonstrated that a single study may significantly impact the overall results (Figure S1). As illustrated in Figure S2, no visual asymmetry was found in the funnel plots for IL-6, CRP, TNF-α, MDA, or 8-isoPGF2α, indicating no evidence of publication bias. There was also no indication of publication bias according to either Begg's rank correlation test or Egger's linear regression test (Table 5). However, the funnel plot and Begg's rank correlation test ($p = 0.060$) for 8-OHdG revealed some publication bias might be present (Figure S2; Table 5).

Discussion

This study included 15 articles from five countries involving 678 subjects and comprehensively assessed the impact of indoor air filtration on biomarkers of inflammation and oxidative stress. The results revealed that indoor air filtration led to a statistically significant reduction in the levels of IL-6. No clear evidence that indoor air filtration can significantly affect the blood concentrations of CRP, TNF-α, MDA, 8-OHdG, and 8-isoPGF2α. Subgroup analysis depicted a significant reduction in 8-OHdG levels in the subgroup with a shorter period of filtration and using a filtrete air filter.

Ambient air pollution is a significant threat to public health. Previous studies have demonstrated that air pollution exposure is closely related to cardiovascular and pulmonary diseases (Yang et al. 2017). An et al. (An et al. 2018) demonstrated that air pollution may increase the instability of atherosclerotic plaque and blood procoagulability and promote the occurrence of cardiovascular events by mediating systemic inflammatory response. Guarnieri et al. (Guarnieri and Balmes 2014) noted that traffic-related air pollution, such as diesel exhaust, is linked to asthma development and exacerbation. Air filters can adsorb and filter various dust, mites, chemicals, and irritants, improve ambient air quality and adjust air humidity (Pui et al. 2008). Many studies have demonstrated that indoor air purifiers can significantly reduce indoor $PM_{2.5}$ concentration, which is potentially beneficial for cardiovascular and respiratory health (Chen et al. 2015; Park et al. 2021). However, the exact mechanism by which indoor air filtration affects health

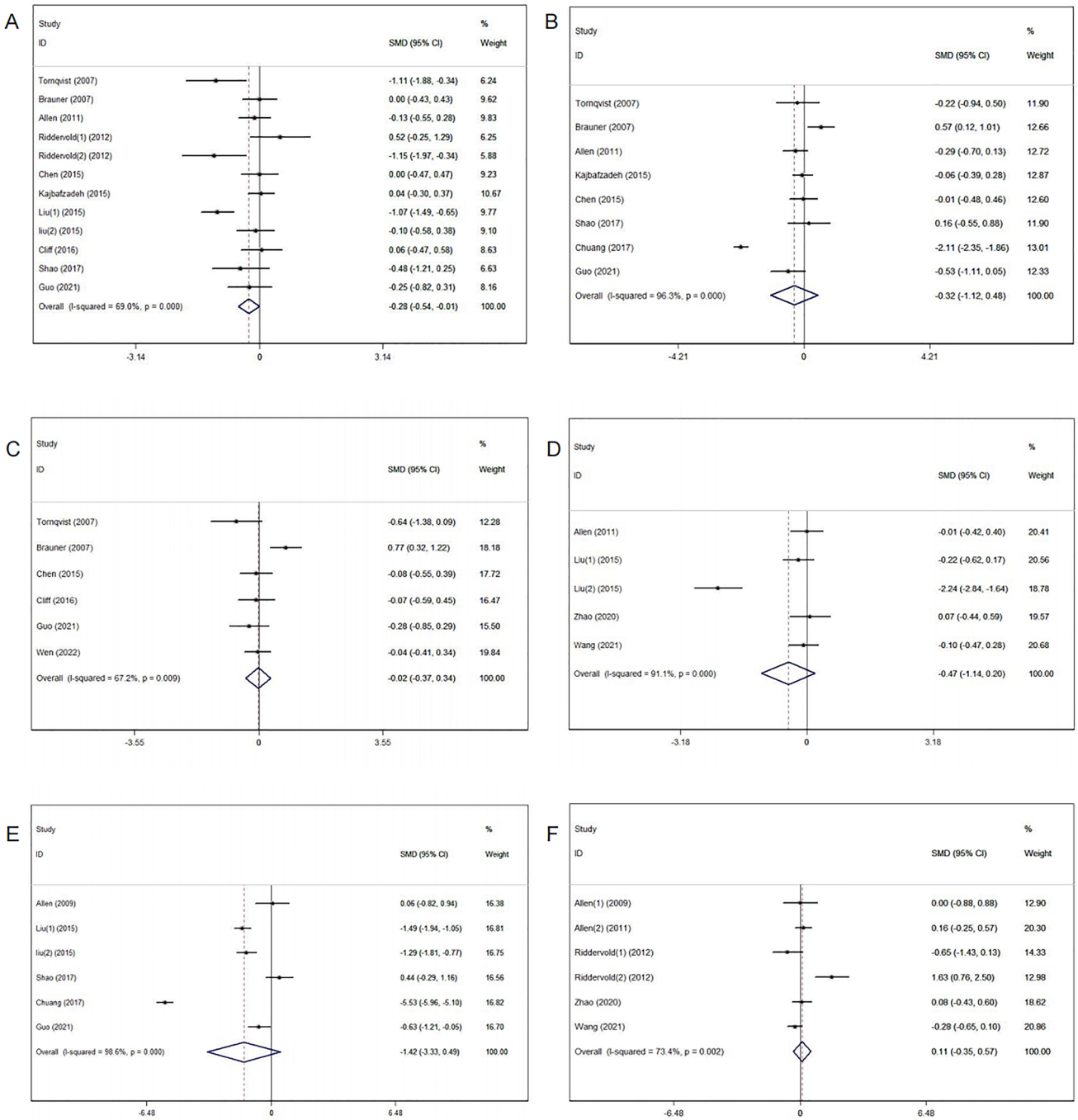


Fig. 2 Forest plot of the effect of indoor air filtration on IL-6 (**A**), CRP (**B**), TNF-α (**C**), MDA (**D**), 8-OHdG (**E**) and 8-isoPGF2α (**F**)

remains unclear. Inflammation and oxidative stress are the key mechanisms in many chronic diseases (Martins et al. 2023). Currently, evidence regarding the impact of indoor air filtration on inflammation and oxidative stress is inconsistent (Zhao et al. 2020). Therefore, we selected IL-6, CRP, TNF-α, MDA, 8-OHdG, and 8-isoPGF2α as outcome indices to explore the relationship between indoor air filtration and biomarkers of inflammation and oxidative stress. To the best of our knowledge, this is the first systematic review and meta-analysis to comprehensively evaluate the effects of indoor air filtration on inflammation and oxidative stress.

Inflammation is also involved in developing various diseases. It is triggered by harmful stimuli and conditions such as infection and tissue damage (Medzhitov 2008). As a pro-inflammatory cytokine, IL-6 is mainly released by T cells and macrophages. This can trigger CRP production in the

Table 3 Assessment of risk of bias in studies included in the meta-analysis

First author (year)	Random sequence generation	Allocation concealment	Blinding of participants	Blinding of outcome assessment	Free of incomplete outcome	Free of selective reporting	Other bias
Törnqvis et al. (2007)	L	L	L	L	L	L	U
Bräuner et al. (2008)	L	L	L	L	H	L	L
Allen et al. (2009)	L	L	L	L	L	L	U
Allen et al. (2011)	L	L	L	L	H	L	U
Riddervold et al. (2012)	L	L	L	L	L	L	U
Kajbafzadeh et al. (2015)	L	L	L	H	H	L	L
Liu et al. (2015)	L	L	L	L	L	L	L
Chen et al. (2015)	L	L	L	L	H	L	U
Cliff et al. (2016)	L	L	L	L	L	L	L
Chuang et al. (2017)	L	L	U	U	L	L	L
Shao et al. (2017)	L	L	U	U	H	L	L
Zhao et al. (2020)	L	L	L	L	L	L	L
Guo et al. (2021)	L	L	L	L	H	L	U
Wang et al. (2021)	L	L	U	L	H	L	L
Wen et al. (2022)	L	L	L	L	H	L	L

L Low risk of bias; *H* High risk of bias; *U* Unclear risk of bias

liver (Kang et al. 2020). Consequently, the concentrations of CRP and IL-6 correlated to some extent. Studies have depicted that serum CRP and IL-6 levels can independently predict cardiovascular events and all-cause mortality (Baune et al. 2011; Shah et al. 2009). TNF-α is an important inflammatory mediator in the inflammatory response (Idriss and Naismith 2000). It can activate neutrophils and lymphocytes, increase the permeability of vascular endothelial cells, regulate the metabolic activities of other tissues, and promote the synthesis and release of other cytokines (Varfolomeev and Vucic 2018). Numerous studies have confirmed that IL-6, TNF-α, and CRP are sensitive indicators of an inflammatory response, which predicts treatment response in patients (Drechsler et al. 2011). Epidemiological data and animal research indicate that inflammatory cytokine levels are higher due to air pollution (Li et al. 2022). In this study, we found that indoor air filtration can significantly reduce IL-6 levels but has no significant effect on TNF-α and CRP levels. Our findings indicate that the underlying mechanisms of the beneficial effects of indoor air filtration on bodily injury may be partly ascribed to inflammation inhibition.

Oxidative stress is closely related to the onset and progression of various diseases. Oxidative stress refers to a state of imbalance between oxidation and antioxidation in the body, easily leading to neutrophil inflammatory infiltration, increased protease secretion, and producing many oxidative intermediates. A large body of evidence indicates that oxidative stress is a negative effect of free radicals in the body and is considered an important factor leading to aging and disease (Hajam et al. 2022). Elevated levels of markers, including MDA, 8-OHdG, and 8-isoPGF2α, are likely to lead to a series of diseases. Available evidence suggests that the formation of excess free radicals is closely related to diet, lifestyle and environmental factors (Sies 2015). Wang et al. (Wang et al. 2021) believed that HEPA air purifiers can reduce the levels of some oxidative stress markers, whereas the other two studies demonstrated that short-term air filtration does not affect oxidative stress (R. W. Allen et al. 2011; Guo et al. 2021). Yu et al. (Yu et al. 2017) believe that $PM_{2.5}$ concentration in the vehicle is closely related to the MDA level of the driver, and a high-efficiency automobile air-conditioning filter can improve the lipid peroxidation of the driver (R. W. Allen et al. 2011). Based on an analysis of published related articles, we selected MDA, 8-OHdG, and 8-isopgf2α as markers of oxidative stress. A meta-analysis demonstrated that indoor air filtration had no significant effect on biomarkers of oxidative stress. When subgroup analysis was performed based on the study duration, short-term indoor air filtration significantly reduced 8-OHdG levels. A possible reason for this difference could be the adherence of participants, since staying indoors for long periods of time is difficult for participants. In addition, subgroup analysis based on filtration technology revealed a significant reduction in 8-OHdG for people who used filtrete air filters. We

Table 4 Subgroup analyses

Variables	Duration		Filtration technology		BMI	
	≥ 1 days	< 1 days	HEPA filter	Filtrete air filter	≥ 24 kg/m^2	< 24 kg/m^2
IL6						
No. Of treatment arms	5	7	7	5	5	4
SMD	-0.051	-0.436	-0.274	-0.299	-0.085	-0.290
(95% CI)	(-0.245 to 0.143)	(-0.901 to 0.030)	(-0.590 to 0.042)	(-0.868 to 0.271)	(-0.285 to 0.114)	(-0.846 to 0.266)
I^2 (%)	0.0	77.0	69.7%	74.3%	0	82.0
P	0.607	0.066	0.090	0.304	0.402	0.306
TNF-α						
No. of treatment arms	3	3	2	4	2	3
SMD	0.212	-0.270	0.260	-0.122	0.260	-0.059
(95% CI)	(-0.317 to 0.742)	(-0.611 to 0.072)	(-0.767 to 1.287)	(-0.364 to 0.120)	(-0.767 to 1.287)	(-0.315 to 0.198)
I^2 (%)	78.0	0.0	87.6	0.0	87.6	0.0
P	0.432	0.121	0.620	0.322	0.620	0.653
CRP						
No. of treatment arms	6	2	5	3	5	2
SMD	-0.300	-0.408	-0.027	-0.796	-0.027	-1.069
(95% CI)	(-1.286 to 0.686)	(-0.857 to 0.042)	(-0.388 to 0.334)	(-2.370 to 0.778)	(-0.388 to 0.334)	(-3.123 to 0.985)
I^2 (%)	97.4	0.0	65.7	97.3	65.7	98.3
P	0.551	0.075	0.883	0.322	0.883	0.308
MDA						
No. of treatment arms	3	2	5	0	1	4
SMD	-0.027	-1.217	-0.473	/	-0.011	-0.599
(95% CI)	(-0.273 to 0.218)	(-3.192 to 0.759)	(-1.144 to 0.197)		(-0.424 to 0.402)	(-1.460 to 0.263)
I^2 (%)	0.0	96.7	91.1	/	/	92.9
P	0.826	0.227	0.166	/	0.958	0.173
8-OHdG						
No. of treatment arms	2	4	5	1	3	3
SMD	-2.555	-0.916	-0.634	-5.530	-0.081	-2.773
(95% CI)	(-8.402 to 3.292)	(-1.513 to -0.320)	(-1.337 to 0.069)	(-5.962 to-5.099)	(-0.762 to 0.601)	(-5.569 to 0.024)
I^2 (%)	99.5	76.4	85.4	/	62.9	99.1
P	0.392	0.003	0.077	<0.001	0.816	0.052
8-isoPGF2α						
No. of treatment arms	3	3	4	2	2	2
SMD	-0.034	0.318	-0.038	0.483	0.130	-0.140
(95% CI)	(-0.316 to 0.248)	(-1.017 to 1.654)	(-0.274 to 0.199)	(-1.753 to 2.719)	(-0.244 to 0.504)	(-0.481 to 0.201)
I^2 (%)	22.8	86.8	0.0	93.2	0.0	17.3
P	0.815	0.640	0.754	0.672	0.496	0.420

IL6 Interleukin-6; *TNF-α* Tumor necrosis factor-α; *CRP* C-reactive protein; *MDA* Malondialdehyde; *8-OHdG* 8-hydroxy-2'-deoxyguanosine; *8-isoPGF2α* 8-iso-prostaglandinF2α

SMD Standardized mean difference; *CI* Confidence intervals

believe that this result may be unconvincing because there was only one study in this subgroup.

In this study, the inflammatory and oxidative stress indicators included were carefully selected and had important clinical significance. Additionally, studies from various countries worldwide were screened and included. Therefore, our research results have reference value for people from different countries and cultural backgrounds. However, we acknowledge the limitations of this study. First, only a small number of studies and patients were included; therefore, the results of some estimations should be cautiously interpreted. Second, several studies included in this review did not have baseline data, which may be an important source of heterogeneity. Third, in terms of the subgroup analysis, each stratum included a relatively small number of studies and patients.

Table 5 Assessment of publication bias

	Begg's rank correlation test			Egger's linear regression test		
	Kendal's score	*Z* value	*P* value	Intercept	95%CI	*P* value
IL6	-16	1.03	0.304	-1.784	(-6.171, 2.604)	0.386
CRP	-2	0.12	0.902	9.741	(-0.660, 20.171)	0.062
TNF-α	-9	1.50	0.133	-4.052	(-13.817, 5.712)	0.313
MDA	-2	0.24	0.806	-13.054	(-35.934, 9.825)	0.167
8-OHdG	11	1.88	0.060	22.074	(-11.934, 56.072)	0.146
8-isoPGF2α	3	0.38	0.707	2.008	(-4.997, 9.012)	0.471

Conclusion

This meta-analysis indicated that indoor air filtration can significantly reduce levels of inflammation and oxidative stress marker in the population. Considering that air pollution affects a wide range of the world and places a significant burden on human health, this discovery highlights the importance of global air pollution control strategies, such as reducing particulate matter, traffic, and industrial emissions, which are of great significance in further improving human health.

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Declarations

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